The HU Collection Part 1

Pure Elegance & Structural Logic

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Abstract

This paper trains candidates for mathematical maturity: the ability to recognize invariants, exploit symmetries, creatively apply geometric principles, and construct elegant arguments, rather than brute forcing. If a solution takes more than a few lines of algebra or requires heavy coordinate bashing, you have missed the correct intuition.

1. Let x, y, and z be real numbers such that $x + y + z = 0$ and $x^2 + y^2 + z^2 = 2$. Evaluate the exact value of:
$$x^4 + y^4 + z^4$$

2. Find the minimum value of the following expression for real x:
$$\sqrt{x^2 + 4} + \sqrt{x^2 - 10x + 29}$$

3. Evaluate the exact value of the following infinite series:
$$\sum_{n=1}^{\infty} \frac{n}{n^4 + 4}$$

4. Determine the number of solutions to the equation:
$$\sin^{2026}(x) + \cos^{2026}(x) = 1$$
in the interval $0 \leq x \leq 2\pi$.

5. Let $f(x)$ be a continuous function defined for all real x such that:
$$f(x) + f(1 - x) = x^2 - x + \frac{1}{2}$$
Evaluate the exact value of the definite integral $\int_0^1 f(x)\,dx$.

6. A whiteboard initially contains the fractions $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots, \frac{1}{2026}$. During each turn, a student chooses two numbers on the board, say a and b, erases them, and replaces them with the single number $a + b - ab$. This process is repeated until exactly one number remains on the board. What is this final number?

7. Find all polynomials $P(x)$ with real coefficients such that for all real x:
$$xP(x - 1) = (x - 3)P(x)$$

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8. 51 points are placed arbitrarily inside a square of side length 1. Prove that there must exist some set of three points among them that can be completely covered by a closed disk of radius $\frac{1}{7}$.

9. Find a cubic equation with integer coefficients whose roots are exactly $\cos\left(\frac{2\pi}{7}\right)$, $\cos\left(\frac{4\pi}{7}\right)$, and $\cos\left(\frac{6\pi}{7}\right)$.

10. Find all real triplets (x, y, z) that satisfy the following system of equations:

$$x + y = z^2$$
$$y + z = x^2$$
$$z + x = y^2$$

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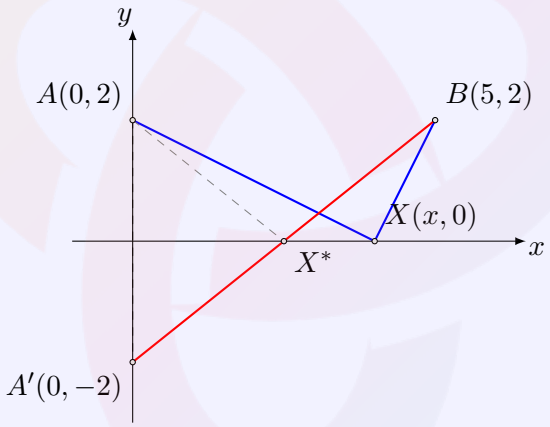
Solutions Appendix

Solution to Question 1

We know $(x+y+z)^2 = x^2+y^2+z^2+2(xy+yz+zx)$. Substituting the given values yield $xy+yz+zx = -1$. Next, we square the sum of squares: $(x^2+y^2+z^2)^2 = x^4+y^4+z^4+2(x^2y^2+y^2z^2+z^2x^2)$. We need to find the value of the cross term. Notice that $(xy+yz+zx)^2 = x^2y^2+y^2z^2+z^2x^2+2xyz(x+y+z)$. Since $x+y+z=0$, $x^2y^2+y^2z^2+z^2x^2=1$. Substituting this back into our squared sum gives $2^2 = x^4+y^4+z^4+2(1)$, thus $x^4+y^4+z^4=2$.

Solution to Question 2

Rewrite the expression as $\sqrt{(x-0)^2+(0-2)^2}+\sqrt{(x-5)^2+(0-2)^2}$. This represents the sum of the distances from a point $X(x,0)$ on the x-axis to the point $A(0,2)$ and the point $B(5,2)$. To minimize this distance, reflect A across the x-axis to $A'(0,-2)$. The shortest path from A' to B is a straight line, which intersects the x-axis at the optimal point X^*. By the distance formula, the minimum value is $A'B = \sqrt{(5-0)^2+(2-(-2))^2} = \sqrt{25+16} = \sqrt{41}$.



Solution to Question 3

Factor the denominator using the Sophie Germain identity: $n^4+4 = (n^2+2)^2-4n^2 = (n^2-2n+2)(n^2+2n+2) = ((n-1)^2+1)((n+1)^2+1)$. We can now use partial fractions to decompose $\frac{n}{n^4+4} = \frac{1}{4}\left(\frac{1}{(n-1)^2+1} - \frac{1}{(n+1)^2+1}\right)$. Let $u_n = \frac{1}{(n-1)^2+1}$. The sum is $\frac{1}{4}\sum_{n=1}^{\infty}(u_n - u_{n+2})$, which is a telescoping series that skips one term. The surviving terms are $\frac{1}{4}(u_1+u_2)$. We calculate $u_1 = \frac{1}{0^2+1} = 1$ and $u_2 = \frac{1}{1^2+1} = \frac{1}{2}$. The sum evaluates to $\frac{1}{4}(1+\frac{1}{2}) = \frac{3}{8}$.

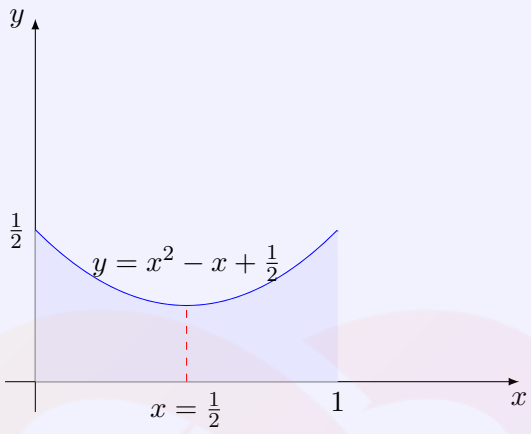
Solution to Question 4

Since $|\sin x| \le 1$ and $|\cos x| \le 1$, elevating them to high even powers makes them smaller or equal. Thus, $\sin^{2026}(x) \le \sin^2(x)$ and $\cos^{2026}(x) \le \cos^2(x)$. Adding them gives $\sin^{2026}(x)+\cos^{2026}(x) \le \sin^2(x)+\cos^2(x) = 1$. For equality to hold, we must have exactly $\sin^{2026}(x) = \sin^2(x)$ and $\cos^{2026}(x) = \cos^2(x)$. This only occurs when $\sin x \in \{0, \pm 1\}$. In the interval $[0, 2\pi]$, this happens at exactly 5 points: $0, \pi/2, \pi, 3\pi/2, 2\pi$.

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Solution to Question 5

Integrate both sides from 0 to 1 to obtain $\int_0^1 f(x)dx + \int_0^1 f(1-x)dx = \int_0^1 (x^2 - x + \frac{1}{2})dx$. Evaluating the right side gives $[\frac{x^3}{3} - \frac{x^2}{2} + \frac{x}{2}]_0^1 = \frac{1}{3} - \frac{1}{2} + \frac{1}{2} = \frac{1}{3}$. By the symmetry of the integral, substituting $u = 1 - x$ reveals that $\int_0^1 f(1-x)dx = \int_0^1 f(x)dx$. Thus, $2\int_0^1 f(x)dx = \frac{1}{3}$, which implies $\int_0^1 f(x)dx = \frac{1}{6}$. The right-hand side curve is symmetric about $x = \frac{1}{2}$, ensuring the relation holds identically over the interval.



Solution to Question 6

Notice that the operation $a+b-ab$ resembles a factorization. Specifically, $1-(a+b-ab) = 1 - a - b + ab = (1-a)(1-b)$. Therefore, if we evaluate the product of $(1-x)$ for all elements x on the board, this product remains invariant after every turn. Initially, the product is: $(1-1/2)(1-1/3)\ldots(1-1/2026) = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \ldots \frac{2025}{2026} = \frac{1}{2026}$. If the final remaining number is F, the invariant dictates that $1 - F = \frac{1}{2026}$, which yields $F = \frac{2025}{2026}$.

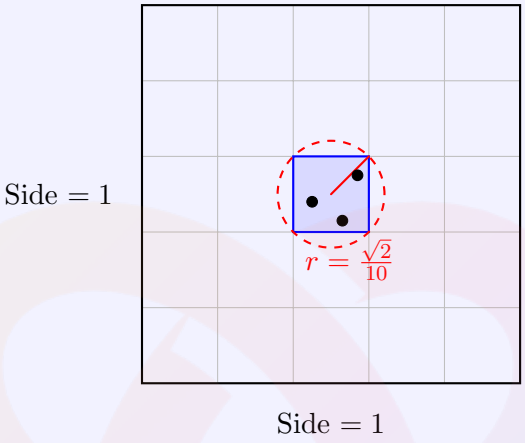
Solution to Question 7

Let $x = 0$. Then $0 \cdot P(-1) = -3 \cdot P(0)$, so $P(0) = 0$. Since $P(0) = 0$, let $x = 1$. Then $1 \cdot P(0) = -2 \cdot P(1)$, so $P(1) = 0$. Since $P(1) = 0$, let $x = 2$. Then $2 \cdot P(1) = -1 \cdot P(2)$, so $P(2) = 0$. Thus, $P(x)$ must contain the factors $x, (x-1)$, and $(x-2)$. Let $P(x) = x(x-1)(x-2)Q(x)$. Substitute this back into the original equation: $x(x-1)(x-2)(x-3)Q(x-1) = (x-3)x(x-1)(x-2)Q(x)$. For $x \notin \{0,1,2,3\}$, we can divide to get $Q(x-1) = Q(x)$. Since a polynomial that is periodic must be constant, $Q(x) = c$. Therefore, $P(x) = cx(x-1)(x-2)$ for any real constant c.

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Solution to Question 8

Divide the 1×1 square into a 5×5 grid of 25 equal, smaller squares. Each small square has a side length of $\frac{1}{5}$. Since there are 51 points and 25 squares, by the Pigeonhole Principle, at least one of these small squares must contain at least $\lceil \frac{51}{25} \rceil = 3$ points.
The maximum distance between any two points in one of these small squares is the length of its diagonal, which is $\frac{\sqrt{2}}{5}$. Furthermore, all points in this square can be covered by the circumcircle of the square. The radius of this circumcircle is half the diagonal: $r = \frac{\sqrt{2}}{10}$.
We must show this circle is smaller than a circle of radius $\frac{1}{7}$. We compare $\frac{\sqrt{2}}{10} \leq \frac{1}{7} \iff 7\sqrt{2} \leq 10 \iff 98 \leq 100$, which is true. Thus, the 3 points fit inside a disk of radius $\frac{1}{7}$.



Solution to Question 9

Let $\zeta = e^{2\pi i/7}$. We know $\zeta^7 = 1$, which factors into $(\zeta-1)(\zeta^6+\zeta^5+\zeta^4+\zeta^3+\zeta^2+\zeta+1) = 0$. Since $\zeta \neq 1$, the degree-6 polynomial equals 0. Divide by ζ^3: $(\zeta^3 + \zeta^{-3}) + (\zeta^2 + \zeta^{-2}) + (\zeta + \zeta^{-1}) + 1 = 0$. Let $y = \zeta + \zeta^{-1} = 2\cos(2\pi/7)$. We can express the powers in terms of y: $\zeta^2 + \zeta^{-2} = y^2 - 2$ and $\zeta^3 + \zeta^{-3} = y^3 - 3y$. Substitute these into the equation: $(y^3 - 3y) + (y^2 - 2) + y + 1 = 0$, giving $y^3 + y^2 - 2y - 1 = 0$. We want the roots to be $x = \cos(\theta)$, so we substitute $y = 2x$: $(2x)^3 + (2x)^2 - 2(2x) - 1 = 0$, which yields $8x^3 + 4x^2 - 4x - 1 = 0$.

Solution to Question 10

Subtract the second equation from the first: $(x+y) - (y+z) = z^2 - x^2$, which simplifies to $x - z = (z-x)(z+x)$. Rearranging yields $(x-z)(1+x+z) = 0$. By cyclic symmetry, we also have $(y-x)(1+y+x) = 0$ and $(z-y)(1+z+y) = 0$. Suppose $x = z$. Substituting back, $x + y = x^2$ and $y + x = x^2$. This means $y = x^2 - x$. The third equation $x + x = y^2$ implies $2x = (x^2 - x)^2 = x^4 - 2x^3 + x^2$. This rearranges to $x^4 - 2x^3 + x^2 - 2x = 0$, factoring into $x(x-2)(x^2+1) = 0$. Thus, $x = 0$ or $x = 2$, yielding the triplets $(0,0,0)$ and $(2,2,2)$. Next, suppose any two variables are equal, say $x = z$, but they do not equal y. Then from $(y-x)(1+y+x) = 0$, since $x \neq y$, $1 + y + x = 0 \implies y = -1 - x$. Substitute this into $y + x = x^2 \implies -1 = x^2$, which has no real solutions. Finally, if all three variables are distinct, then $1 + x + z = 0$ and $1 + y + x = 0$, which implies $z = y$, a contradiction. Thus, the only real solutions are $(0,0,0)$ and $(2,2,2)$.

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